



Introduction to Photogrammetry

Lecture 02

Introduction, History & Importance of Photogrammetry

Usman Mazhar, Ph.D.

usmanmazhargeo@outlook.com

In Today's Class

- What is Photogrammetry?
- History of Photogrammetry
- Why Do We Need Photogrammetry?
- Applications of Photogrammetry
- Photogrammetry and remote sensing
- Photogrammetry and Surveying
- Types of Photogrammetry

1. What is Photogrammetry

- Photogrammetry is the science and technology of obtaining reliable measurements and information about objects from photographs.
- Break the word:
- Photo = Light
- Gram = Drawing or recording
- Metry = Measurement
- So, photogrammetry means measurement from photographs.

- Photogrammetry is the science of obtaining reliable information about the properties of surfaces and objects without physical contact with the objects, and of measuring and interpreting this information.
- Purpose: It focuses on the geometric, quantitative, and spatial attributes of features, such as height, volume, and location.
- Process: It involves acquiring overlapping photographs (stereo imagery) from sensors (drones, aircraft, satellites) and using specialized software to process them into 3D models.

History of Photogrammetry

- The development of photogrammetry clearly depends on the general development of science and technology.
- It is interesting to note that the four major phases of photogrammetry are directly related to the technological inventions of photography, airplanes, computers and electronics.

First generation

- Photogrammetry had its beginning with the invention of photography by Daguerre and Niepce in 1839.
- The first generation, from the middle to the end of last century, was very much a pioneering and experimental phase with remarkable achievements in terrestrial and balloon Photogrammetry

Second generation

- The second generation, usually referred to as analog photogrammetry, is characterized by the invention of stereophotogrammetry by Pulfrich (1901).
- This paved the way for the construction of the first stereoplotter by Orel, in 1908.
- Airplanes and cameras became operational during the first world war.
- Between the two world wars, the main foundations of aerial survey techniques were built and they stand until today.

- Analog rectification and stereoplotting instruments, based on mechanical and optical technology, became widely available.
- Photogrammetry established itself as an efficient surveying and mapping method.
- The basic mathematical theory was known, but the amount of computation was prohibitive for numerical solutions and consequently all the efforts were aimed toward analog methods.
- Von Gruber is said to have called photogrammetry the art of avoiding computations.

Third generation

- With the advent of the computer, the third generation has begun, under the motto of analytical photogrammetry.
- Schmid was one of the first photogrammetrists who had access to a computer.
- He developed the basis of analytical photogrammetry in the fifties, using matrix algebra.
- For the first time a serious attempt was made to employ adjustment theory to photogrammetric measurements. It still took several years before the first operational computer programs became available.

- Brown developed the first block adjustment program based on bundles in the late sixties, shortly before Ackermann reported on a program with independent models as the underlying concept.
- As a result, the accuracy performance of aerial triangulation improved by a factor of ten.
- Apart from aerial triangulation, the analytical plotter is another major invention of the third generation.
- Again, we observe a time lag between invention and introduction to the photogrammetric practice. Helava invented the analytical plotter in the late fifties.
- However, the first instruments became only available in the seventies on a broad base.

Fourth generation

- The fourth generation, digital photogrammetry, is rapidly emerging as a new discipline in photogrammetry.
- In contrast to all other phases, digital images are used instead of aerial photographs.
- With the availability of storage devices which permit rapid access to digital imagery, and special microprocessor chips, digital photogrammetry began in earnest only a few years ago.
- The field is still in its infancy and has not yet made its way into the photogrammetric practice.

Why Do We Need Photogrammetry?

- Photogrammetry is essential for creating accurate 3D models and digital maps from 2D photographs, providing a fast, cost-effective, and safe alternative to traditional ground surveying. It is critical for industries like construction, archaeology, and mining, allowing for precise measurements, volumetric calculations, and detailed documentation of sites or objects.

- **Efficiency in Data Acquisition:** It enables the rapid scanning of large or complex sites, including inaccessible areas, to create detailed 3D models.
- **High Accuracy for Measurement:** It is used to generate accurate topographic maps, point clouds, and 3D models, which are vital for engineering, planning, and volume estimation.
- **Versatility Across Industries:** Construction & Engineering, Cultural Heritage & Archaeology, Forensics.
- **Cost-Effectiveness:** Using drone-based, aerial, or ground-level photography reduces the need for expensive manual surveying equipment and labor.

Applications of Photogrammetry in Pakistan

- **Surveying and Mapping:** Creating topographic maps, digital elevation models (DEMs), and digital surface models (DSMs) for engineering projects, land planning, and cadastral mapping.
- **Construction and Infrastructure:** Inspecting infrastructure such as telecommunication towers, wind turbines, and bridges, as well as calculating earthwork volumes and monitoring site progress.
- **Archaeology and Cultural Heritage:** Documenting, mapping, and creating 3D replicas of archaeological sites, artifacts, and historical structures.

- Drone/UAV Mapping: Using drones to capture high-resolution imagery for real-time surveying, urban planning, and resource management.
- Environmental Monitoring: Analyzing flood zones, coastal erosion, deforestation, and geological formations.
- Film and Gaming: Creating 3D digital assets, realistic textures, and environmental models for CGI, video games, and virtual reality.

- Forensics and Inspection: Documenting accident scenes, forensic evidence, and non-visible details using specialized sensors.
- Agriculture: Analyzing crop health and optimizing land use through aerial surveillance.

Photogrammetry and remote sensing

- Photogrammetry and remote sensing are two related fields.
- The International Society of Photogrammetry and Remote Sensing (ISPRS) is a non-governmental organization devoted to the advancement of photogrammetry and remote sensing and their applications.
- The principal difference between photogrammetry and remote sensing is in the application

- while photogrammetrists produce maps and precise three-dimensional positions of points,
- remote sensing specialists analyze and interpret images for deriving information about the earth's land, atmosphere and water areas.
- Both disciplines are also related to Geographic Information Systems (GIS) in that they provide GIS with essential information.
- The core of topographic information is produced by photogrammetrists in form of a digital map.

Surveying & Photogrammetry

Surveying

Field-based measurement

Uses total station

Small area

Explain in simple terms.

Photogrammetry

Image-based measurement

Uses aerial photos or satellite images

Large area

Types of Photogrammetry

- Terrestrial Photogrammetry
 - Uses ground cameras for detailed 3D modeling of objects or buildings.
- Aerial Photogrammetry
 - Uses drones, or planes, for large-scale mapping
- Satellite Photogrammetry
 - Uses satellite for large-scale mapping



Thank you!

Any Questions
